Machine learning for predicting sediment particle size distributions

UE24CS352A Machine Learning · Mini-project 62 · Team B-16 · Bandugula Sai Rithesh Reddy (PES1UG24AM064), Gagana C (PES1UG24AM099)

1. Problem statement

Mud particles suspended in estuaries stick together into flocs and break apart again within seconds to minutes. Their size sets how fast they settle, which is the most uncertain input to sediment-transport, erosion and contaminant models. No general physical model predicts floc size well. Following Egan (Stanford CS229, 2019), we ask whether a regression model can predict two summaries of the particle size distribution (PSD) from routinely measured water properties: the **median diameter d50** and the **spread σ** (the volume-weighted standard deviation of diameter, i.e. the square root of the paper's variance σ^2). Success is measured, as in the paper, by the coefficient of determination R^2 on held-out data.

2. Dataset

The paper's own data are not public, so we used the closest public equivalent: the U.S. Geological Survey release Profiles of salinity, temperature, depth, turbidity, and distributions of particle size in suspension … south San Francisco Bay, summer 2020 (doi:10.5066/P99Q4CHM). It covers the same bay with the same instrument family (a Sequoia LISST-200X laser particle sizer).

- **Coverage:** vertical casts at a deep channel site (11–14 m) and a shallows site (2–5 m) on four days (21, 22, 24, 28 July 2020). The package was held about a minute at each depth.
- **Raw size:** 94 NetCDF files, 45,639 one-second samples after removing one file that is a duplicate of another. Each sample has the volume concentration in 36 size classes (1.2–459 µm), depth, temperature and beam attenuation.
- **After cleaning:** 3,731 ten-second bursts from 89 casts.
- **What is missing:** the CTD files holding salinity and turbidity could not be downloaded (the published links return “page not found”), and the release has no chlorophyll or refractive index. Salinity was the paper's strongest predictor, so our feature set is weaker than theirs.

3. Approach

Targets. We recompute d50 and σ from each burst-averaged PSD; our d50 agrees with the value stored in the files ($r = 0.97$). **Features (7).** Sensor depth, height above the bed, total water depth (the tidal stage), its rate of change between casts (flood or ebb), temperature, beam attenuation (a proxy for sediment concentration) and a site flag. Total volume concentration was deliberately left out because it is computed from the PSD itself.

Models. We reproduce the paper's two models, Random Forest and support vector regression, and add a ridge baseline, gradient boosting and a 10-unit neural network (the architecture of Sahin et al., 2017). As in the paper, feature importance is averaged over repeated 10-tree forests, the forest size (50 trees) is chosen where the out-of-bag score levels off, and SVR's C and ε come from a validation sweep ($C = 0.25$, $\varepsilon = 0.03125$ on standardised targets).

Evaluation: the main extension. The paper closes by asking whether its models would work at another time or place. Consecutive samples in a cast are nearly identical, so a random split puts near-copies of every test point in the training set. We therefore report four protocols of increasing difficulty: (1) the paper's random 70/30 split, (2) five-fold cross-validation with whole casts held out, (3) leave-one-day-out, and (4) train on one site, test on the other.

4. Implementation

- **src/build_dataset.py** reads the NetCDF casts, decodes the EPIC time format, drops in-air samples, samples outside 30–98 % optical transmission (the instrument's valid range) and readings below the sea bed, then averages into 10-second bursts taken at a steady depth.
- **src/psd.py** computes d50 (log-interpolated on the cumulative volume curve) and σ.
- **src/train_evaluate.py** runs feature importance, tuning, all five models under all four protocols and a feature-set ablation; **src/plots.py** draws the figures.
- **demo.py** holds out one cast, retrains and prints predicted against measured values, or predicts for conditions typed on the command line. Python 3, scikit-learn, xarray; fixed seed.

5. Results

R^2 on test data for each model and protocol (mean over splits; higher is better, below 0 is worse than predicting the mean).

Model	Random 70/30 (paper)		Unseen casts		Unseen day		Unseen site	
	d50	σ	d50	σ	d50	σ	d50	σ
Ridge (linear baseline)	0.38	0.42	0.32	0.39	0.11	0.17	< −10	< −10
Random Forest	0.84	0.78	0.41	0.54	0.32	0.42	-1.05	-0.93
SVR (RBF)	0.64	0.64	0.52	0.58	0.46	0.47	-1.75	-2.03
Gradient Boosting	0.83	0.77	0.46	0.55	0.42	0.45	-1.48	-0.83
Neural net (10 units)	0.67	0.65	0.51	0.55	0.43	0.40	< −10	< −10

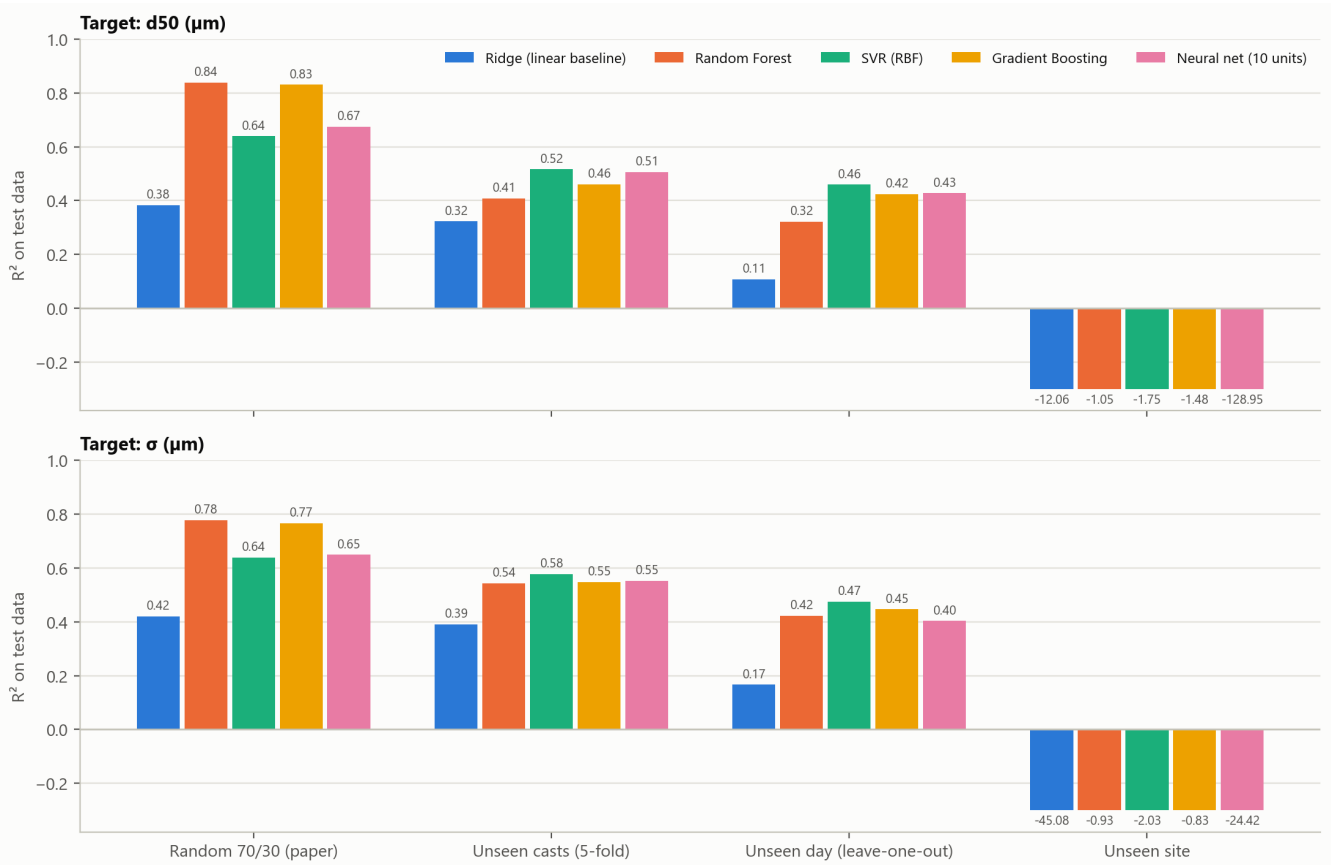


Figure 1. R^2 of every model under the four test protocols. Bars are cut off at −0.3; labels give the true value.

- **The paper's result reproduces.** With a random split, Random Forest reaches $R^2 = 0.84$ for d50 and 0.78 for σ; the paper reports 0.81 and 0.75 on different data and features.
- **Most of that score is memorised neighbours.** On casts the model never saw, Random Forest falls to 0.41 and 0.54. SVR, the smoother model, is now best (0.52 and 0.58); on an unseen day it scores 0.46 and 0.47.
- **Nothing transfers between sites.** Every model has negative R^2 when trained on one site and tested on the other, because water depth and particle populations barely overlap.
- **What drives the predictions.** Water depth (tide stage), sensor depth and beam attenuation matter most; site and tide rate add almost nothing. Position, tide and temperature alone give $R^2 = 0.44$ (d50) and 0.47 (σ) on unseen casts with Random Forest; adding attenuation helps σ but not d50.

6. Conclusions

Machine learning can explain roughly half of the variation in floc size and spread on new casts from depth, tide, temperature and optical attenuation alone, which is useful but well short of the 0.75–0.84 a random split suggests. The headline lesson is methodological: for autocorrelated field data the split decides the score, and the paper's worry about overfitting was justified. Limits of our study: four days in one summer, two sites, and no salinity or turbidity. Adding the CTD data and the hydrodynamic measurements from the parent USGS release is the obvious next step; the pipeline merges CTD casts automatically when they are present, though that path is untested.